



King Abdulaziz University

Faculty of Computing & Information Technology Computer Science Department

Exam 1 (Thursday October 10, 2019)
CPCS-223 (Analysis and Design of Algorithms)
Fall 2019
Points: 60 Time: 80 minutes

SOLUTION

Student ID: _____

Student Name: _____

***Section: _____ GA

Q	SO	Question	Max. Marks	Obt. Marks
01	A	1	10	
02	A	2	10	
03	A	3	10	
04	B	4	10	
05	B	5	10	
06	A	6	10	

Instructor:

☐ Dr Muhammad Umair

CLO – 1
Question No. 1

SO – A
(01 + 09) points

How we may specify an “Algorithm”? Write down its names. Elaborate the difference between them with one example.

Solution:

1. Natural Language
2. Code Segments
3. Pseudo Code

- **Specifying an Algorithm (Natural Language)**

Step 1: Take input an integer “a”
Step 2: Take input an integer “b”
Step 3: Add “a” and “b”
Step 4: Store result in “c”
Step 5: Display “c”

- **Specifying an Algorithm (Code Segment)**

- Scanner in = new Scanner(System.in);
- a = in.nextInt();
- b = in.nextInt();
- c = a + b;
- System.out.println(c);

- **Specifying an Algorithm (Pseudo Code)**

- Take input of two values “a” and “b”
- c = a + b;
- Display the value “c”

CLO – 2

SO – A

Question No. 2

10 points

On which parameters the efficiency of an algorithm depends? Describe the main characteristics of “Space Efficiency” for an algorithm.

Solution:

Efficiency Parameters:

(02) pts

1)

Time or Speed

to determine the performance of the algorithm in terms of **time taken (Time Efficiency)**

2

Space or Memory

to determine the performance of the algorithm in terms of **storage requirements (Space Efficiency)**

Space Efficiency:

(08) pts

S. No.	Characteristics
1	The space efficiency determines total memory requirement of RAM and disk storage to run an algorithm with given input.
2	The space requirement consists of the amount of real storage needed to execute the algorithm code and the storage to hold the application data.
3	The algorithm code occupies a fixed amount of space, which is independent of the input size.
4	The storage requirement for the application depends on the nature of data structure used to provide faster and flexible access to stored information.
	some algorithms require extra storage to hold results of intermediate (not final i.e. prior to final) computations.

PART – A

Prove that

$$(3/2) n^2 - 4n = \Theta (n^2)$$

Solution:

We have to prove that $c_2 (n^2) \leq (3/2) n^2 - 4n \leq c_1 (n^2)$

It will be solved in two steps;

Step -1

$$(3/2) n^2 - 4n \leq c_1 (n^2)$$

Step -2

$$(3/2) n^2 - 4n \geq c_2 (n^2)$$

Step – 1

$$(3/2) n^2 - 4n \leq c_1 (n^2)$$

$$(3/2) - 4/n \leq c_1$$

For $c_1 \geq 3/2$ and $n_0 = 1$, the relation holds

Step – 2

$$(3/2) n^2 - 4n \geq c_2 (n^2)$$

$$(3/2) - 4/n \geq c_2$$

For $n = 1$

$$(3/2) - 4/n = (3/2) - 4 = -5/2$$

For $n = 2$

$$(3/2) - 4/n = (3/2) - 2 = -1/2$$

For $n = 3$

$$(3/2) - 4/n = (3/2) - 4/3 = 1/6$$

So for $c_2 = 1/6$ and $n_0 = 3$, the relation holds

Hence proved

$$(3/2) n^2 - 4n = \Theta (n^2)$$

PART – B

Solve the given recurrence using “Backwards Substitution Method”

$$T(n) = T(n - 1) + n$$

Solution:

$$T(n) = T(n-1) + n$$

Put $n = n-1$ we get $T(n-1) = T(n-2) + (n-1)$, replace this value

$$T(n) = T(n-2) + (n-1) + n$$

Put $n = n-2$ we get $T(n-2) = T(n-3) + (n-2)$, replace this value

$$T(n) = T(n-3) + (n-2) + (n-1) + n$$

Put $n = n-3$ we get $T(n-3) = T(n-4) + (n-3)$, replace this value

$$T(n) = T(n-4) + (n-3) + (n-2) + (n-1) + n$$

.....

.....

.....

$$T(n) = T(3) + 4 + 5 + 6 + \dots + (n-3) + (n-2) + (n-1) + n$$

Put $n = 3$ we get $T(3) = T(2) + 3$, replace this value

$$T(n) = T(2) + 3 + 4 + 5 + 6 + \dots + (n-3) + (n-2) + (n-1) + n$$

Put $n = 2$ we get $T(2) = T(1) + 2$, replace this value

$$T(n) = T(1) + 2 + 3 + 4 + 5 + 6 + \dots + (n-3) + (n-2) + (n-1) + n$$

Put $n = 1$ we get $T(1) = 1$, replace this value

$$T(n) = 1 + 2 + 3 + 4 + 5 + 6 + \dots + (n-3) + (n-2) + (n-1) + n$$

$$T(n) = n(n+1)/2 = \frac{1}{2} n^2 + \frac{1}{2} n = \Theta(n^2)$$

CLO – 4

SO – B

Question No. 4

10 points

Find the running time of the following algorithm using the RAM model and approximate the running time with Θ – notation.

ALGORITHM *UniqueElements*($A[0..n - 1]$)

```

for  $i \leftarrow 0$  to  $n - 2$  do
    for  $j \leftarrow i + 1$  to  $n - 1$  do
        if  $A[i] = A[j]$  return false
return true

```

Solution:

(04 pts)

Statement #	Cost	Times(# of Iterations)
<u>1</u>	C1	n
<u>2</u>	C2	$\sum_{i=0}^{n-2} (n - 1 - i)$
<u>3</u>	C3	$\sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1$
<u>4</u>	C4	1

T (n) =

(06 pts)

$$C_{worst}(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1$$

$$= \sum_{i=0}^{n-2} [(n - 1) - (i + 1) + 1]$$

$$= \sum_{i=0}^{n-2} (n - 1 - i)$$

$$= \sum_{i=0}^{n-2} (n - 1) - \sum_{i=0}^{n-2} i$$

$$\begin{aligned}
C_{worst}(n) &= \sum_{i=0}^{n-2} (n-1) - \sum_{i=0}^{n-2} i \\
&= (n-1) \sum_{i=0}^{n-2} 1 - \frac{(n-2)(n-1)}{2} \\
&= (n-1)^2 - \frac{(n-2)(n-1)}{2} \\
&= \frac{(n-1)n}{2} \\
&\in O(n^2)
\end{aligned}$$

CLO – 5**Question No. 5**

For the given data set, how the arrays A will be updated during execution of the given algorithm of “Question05”? You are required to reflect all changes to these arrays for each iteration.

SO – B**10 points**

```

INSERTION-SORT(A)
1  for  $j \leftarrow 2$  to  $\text{length}[A]$ 
2      do  $\text{key} \leftarrow A[j]$ 
3          ▷ Insert  $A[j]$  into the sorted sequence  $A[1 \dots j - 1]$ .
4           $i \leftarrow j - 1$ 
5          while  $i > 0$  and  $A[i] > \text{key}$ 
6              do  $A[i + 1] \leftarrow A[i]$ 
7               $i \leftarrow i - 1$ 
8           $A[i + 1] \leftarrow \text{key}$ 

```

Solution:**A**

0	1	2	3	4	5	6	7	8	9
49	68	05	61	35	57	6	44	99	91

Iteration	0	1	2	3	4	5	6	7	8	9
1	49	68	05	61	35	57	6	44	99	91
2	05	49	68	61	35	57	6	44	99	91
3	05	49	61	68	35	57	6	44	99	91
4	05	35	49	61	68	57	6	44	99	91
5	05	35	49	57	61	68	6	44	99	91
6	05	6	35	49	57	61	68	44	99	91
7	05	6	35	44	49	57	61	68	99	91
8	05	6	35	44	49	57	61	68	99	91
9	05	6	35	44	49	57	61	68	91	99

CLO – 6
Question No. 6

SO – A
(10) points

Write the pseudo-code / algorithm to calculate the sum of all digits in a given number.

Input: a number from user e.g. 348765

Process: separate the digits of the number and add them up

Output: return the sum value 33

Solution:

```
public static int sumDigits(int n) {  
    if ( n == 0 )  
        return 0;  
    else  
        return sumDigits(n/10) + (n%10);  
}
```

Extra Sheet for rough work